

SORA: SCALABLE BLACK-BOX REACHABILITY ANALYSER ON NEURAL NETWORKS

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ABSTRACT

The vulnerability of deep neural networks (DNNs) to input perturbations has posed a significant challenge. Recent work on robustness verification of DNNs not only lacks scalability but also requires severe restrictions on the architecture (layers, activation functions, etc.). To address these limitations, we propose a novel framework, SORA, for scalable blackbox reachability analysis of DNNs. SORA can work on a broad class of neural network structures, including those networks with very deep layers and a huge number of neurons with non-linear activation functions. Based on the Lipschitz continuity, SORA verifies the reachability property of DNNs with a novel optimisation algorithm and has global convergence guarantee. Our method does not require access to the inner structures of the DNNs, hence a black-box method. Experimental results show that, compared to existing verification methods, SORA shows superior performance in terms of both efficiency and scalability, especially when handling a deep neural network with very deep layers and a large number of neurons with various types of nonlinear activation functions.

Index Terms— Safe, Reachability, Lipschitz, Scalability

1. INTRODUCTION

DNNs are surprisingly vulnerable to imperceptible adversarial perturbations [1, 2, 3]. Although numerous studies have analysed the adversarial robustness of DNNs, most of these investigations cannot provide safety guarantees for a given network. Taking the image classification task as an example, only a few verification methods can determine whether the output of a classifier can be altered by a certain amount of perturbation inside an example with theoretical guarantees [4, 5, 6, 7, 8]. To provide a provable robustness of DNNs, [9, 10] reduces the robustness verification problem to a constraint satisfaction problem. Singh *et al* [11, 12] and Wang *et al* [13] propagate a pointwise relaxation through a network to capture the output range of its final layer. Search-based approaches [14, 15] discretise the neighbourhood of an input and perform an exhaustive search. However, most existing verification methods are unscalable. They can only verify tiny networks with limited activation operations (*e.g.* ReLU

activation function or feedforward layers), which starkly contrasts with industrial-scale DNNs that usually have millions or even billions of neurons. Moreover, most verification methods [16, 17, 18, 19, 20, 21, 22] require accessing details of target models, such as model parameters and first-order gradient, which is inapplicable and inefficient in practise.

In response to the limitations of current verification methods, this paper proposes a generic and efficient method, SORA, which is capable of verifying large-scale DNNs in a black-box environment without access to model structures and parameters. SORA is motivated by recent advances in Lipschitz optimisation [23]. It only requires the target model to be Lipschitz continuous and is not limited by the model's scales and architectures. Given an input and an objective function, SORA repeatedly divides the space and identifies potential optimal subspaces that are likely to contain the global optimal solution for the next partition. Theoretically, SORA is guaranteed to converge within a finite number of queries to the global optimal point, which is desirable for robustness verification tasks. Therefore, we implement SORA in a GPU-friendly way and adopt it to perform reachability analysis and safety verification on various types of neural networks. Specifically, the contributions are as follows:

- We propose a global optimisation based robustness analyser, SORA, which is guaranteed to locate the optimal subspace that contains the global minimum of the target model within a finite number of queries. Unlike DeepGO[24], SORA does not require the Lipschitz constant of target models and can handle models with higher input dimensions.
- Different from other white-box methods (*e.g.*, ERAN [12], crown [13]), SORA is a black-box reachability analyser which can work on any Lipschitz continuous DNNs.
- SORA outperforms the existing reachability methods in terms of both efficiency and tightness. SORA is *ten times faster* than $\alpha\beta$ -crown [13] to find a comparable bound for the maximum safe norm ball.

2. METHODOLOGY

2.1. Problem Formulation

Given a Lipschitz continuous N-layer neural network $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with input space $\mathcal{X} \subset \mathbb{R}^n$ and output space

PX conducted this work while she was visiting the University of Exeter. PX and FW contributed to this work equally.

$f(x) = \{c_1, \dots, c_m\}$, that satisfies $\|f(x_1) - f(x_2)\|_a \leq K\|x_1 - x_2\|_b, \forall x_1, x_2 \in \mathcal{X}$, where c_i is the confidence value, K is Lipschitz constant, $\|\cdot\|_a$ and $\|\cdot\|_b$ are two metrics. We aim to study the maximum safe norm ball (MSNB) of the target DNN model f towards a given example $x \in \mathbb{R}^n$.

Definition 1 (Maximum Safe Norm Ball (MSNB)) Given a pair of input example and label (x, y) and a norm ball $\mathcal{B}(x, d, p) = \{x' \mid \|x' - x\|_p \leq d\}$, a DNN f is safe on example x if and only if

$$\forall x' \in \mathcal{B}(x, d, p) : \arg \max_{j=1, \dots, m} f(x') = y. \quad (1)$$

where y is original label of the clean image. When d obtains the maximal d_{max} , we call it as the maximum safe norm ball $\mathcal{B}(x, d_{max}, p)$.

Computing the MSNB is essentially a safety verification problem [14] that can be reduced to solving reachability problems. To measure the difference of confidence values of original label and largest confidence values of the other labels, let $g = c_y - \max_{i \neq y} (c_1, \dots, c_m)$ with the definition of reachable set as follows,

Definition 2 (Reachable Set) For a DNN $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, let $X' \subseteq [0, 1]^n$ be an input subspace, the output reachable set of DNN f over the function g can be defined as:

$$\mathcal{R}(g, X') \triangleq \{\min_{x' \in X'} g(f(x')), \max_{x' \in X'} g(f(x'))\} = \{l, u\} \quad (2)$$

where $l \triangleq l(g, X') = \min_{x' \in X'} g(f(x'))$ and $u \triangleq u(g, X') = \max_{x' \in X'} g(f(x'))$ are lower and upper bound, respectively.

If $g > 0$, then the neural network is safe with the lower bound $l = \min_x g(f(x)) = 0$. Otherwise, the neural network is unsafe. We can verify the safety of a DNN f with respect to x and $\mathcal{B}(x, d, p)$ if and only if $\min g(f(x')) \geq 0, \forall x' \in \mathcal{B}(x, d, p)$ holds. That is, the safety problem is reduced to check if 0 is within the reachable set.

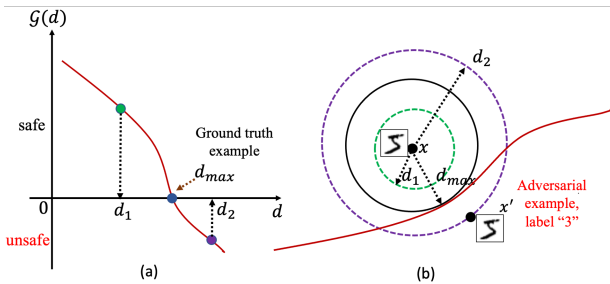


Fig. 1. A schematic figure showing maximum safe radius d

We compute the maximum safe radius as the maximum d on which the value 0 is not reachable for the composite function $g \cdot f$ on the norm ball $\mathcal{B}(x, d, p)$. Generally, we define the function $\mathcal{G}$ with respect to the radius d as $\mathcal{G}(d) = \min_{x' \in \mathcal{B}(x, d, p)} g(f(x'))$. When using binary search

to increase the radius d from smaller values to larger ones, the output of $\mathcal{G}(d)$ will decrease continuously and monotonically from positive to negative. According to the intermediate value theorem [25], there exists a radius $d' \in [0, 1]$, which takes a value 0, i.e. $\mathcal{G}(d') = 0$. We indicate it as the maximum safe radius d_{max} . When $d_1 < d_{max}$ represents that the network is safe within d , otherwise the network is unsafe with respect to $d_2 > d_{max}$, as shown in Fig. 1(a).

In Fig. 1(b), the green dashed line represents a lower bound of the maximum safe norm radius. Noted that robustness verification or certification in DNNs mainly aims to find a certified lower bound as tight as possible to the d_{max} [26, 27]. The dashed violet line means the upper bound of the maximum safe norm radius, and adversarial attacks can be used to generate the upper bound of d_{max} [5, 28, 4]. Thus, by instantiating the function g , we can quantify the output/logit range of a DNN and verify whether a DNN is safe.

In summary, the goal of reachability analysis on DNN is to solve the following problem:

$$\min_x w(x) = g \cdot f, \quad s.t. \quad x \in [a, b]^n \quad (3)$$

Due to the highly non-convex and non-differential nature of DNNs, the exact computation of the global minimum of the composite function $w(x)$ is a very challenging problem [9, 29, 10]. Moreover, regular optimisation approaches based on gradient descent cannot achieve the global minimum with guaranteed convergence [1, 28]. As $w(x)$ is a black-box function that is costly to evaluate. Recently, some specific optimisation algorithms have been proposed to address this problem. One classic optimisation method is based on Lipschitz optimisation. Existing research [24] presented a novel algorithm based on adaptive nested optimisation to solve the reachability problem. However, there are some limitations in [24]. One is that the algorithm needs the Lipschitz constant. Another problem lies in the initialisation phase. Although the number of function evaluations is linear at the endpoint of the intervals, the n dimensional evaluation of the algorithm quickly becomes intractable, which amounts to 2^n exponential time. To alleviate these issues, we explore a new algorithm based on the Lipschitzian optimisation of dividing rectangles [23], named SORA, which can solve this problem more efficiently and practically with soundness.

2.2. Reachability Analysis via Lipschitzian Optimisation

The *only* requirement to apply SORA is that the target neural networks for verification are Lipschitz continuous. The authors in [1, 24] demonstrated deep neural networks with convolutional, max-pooling layer and fully-connected layers with ReLU, Sigmoid, Hyperbolic Tangent activation, and Softmax layer are Lipschitz continuous. Here we can prove that Convolutional or fully-connected layers with GELU (Gaussian error linear unit) activation function

$(Wx + b)\Phi(Wx + b)$ (where $\Phi(\cdot)$ is a cumulative distribution function), Softplus $S(Wx + b)$ and PReLU activation function are Lipschitz continuous and their Lipschitz constants are $(nC\|W\| + \|b\|)\|W\|$, $\|W\|$ and $\max\{\alpha, 1\}\|W\|$ respectively. The proof can be derived based on the definition of Lipschitz continuity and composition operator [30], which we omit here due to the page limit.

As a deterministic splitting algorithm, SORA can optimise the problem Eqn. (3) by the subdivision of input space. The procedure includes sequential queries of the unknown function $w(x)$ with input space division. When *trisection* n -dimensional hyperrectangle, there includes m dimensions with long side length $\sigma_i = 3^{-l}$ and $n - m$ dimensions with short side length $\sigma_j = 3^{-l-1}$, where l represents the division level. We query the value of the function at the centre points $\mathbf{c} \pm 3^{-l}\mathbf{e}_i$ and subdivide the space into subrectangles according to the best query function values, which are given by $\min\{w(\mathbf{c} + 3^{-l}\mathbf{e}_i), w(\mathbf{c} - 3^{-l}\mathbf{e}_i)\}$. We visualize the trisection for 1 Dimensional and 2 Dimensional function in Fig. 2. After subspace division, we need to identify the new potentially optimal subspace for the next division using the following conditions. We repeat this process until the stop criteria are satisfied.

$$w(\mathbf{c}_j) - \tilde{K}\sigma_j \leq w(\mathbf{c}_i) - \tilde{K}\sigma_i, \forall i \in \{1, 2, \dots, n\} \quad (4)$$

$$w(\mathbf{c}_j) - \tilde{K}\sigma_j \leq w_{\min} - \epsilon|w_{\min}| \quad (5)$$

For Eqn. (4) shows the biggest potential improvement of the current minimum $w_{\min}$. The second condition (5) guarantees the improvement greater than $\epsilon|w_{\min}|$, which is not a trivial improvement. The reasonable choice of ϵ is between 10^{-3} – 10^{-7} in [23].

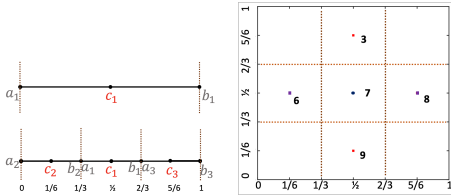


Fig. 2. Trisection Examples: (a) 1-D function; (b) 2-D function

SORA can achieve the minimal value without fixed Lipschitz constant K . To illustrate more clearly, we use the distance σ_j as x-axis and function $w(c_j)$ as y-axis and obtain some dots in Fig. 3. When moving a line with slope K upward until it passes one blue dot and leaves all other dots above it, we can obtain the vertical intercept as the lower bound $lb(\sigma_j) = w(c_j) - K\sigma_j$. However, using only one constant K as slope generally has a high emphasis on global search and leads to slow convergence because the Lipschitz constant is the upper bound on the rate of global search ($w(\mathbf{c}_j)$) versus local search (σ_j). To avoid a single and large Lipschitz constant, we consider all relative weightings to identify the set of intervals, such as the red dashed line $\tilde{K}$ in Fig. 3. As can be observed in Fig. 3(b), the dots located on the lower right part

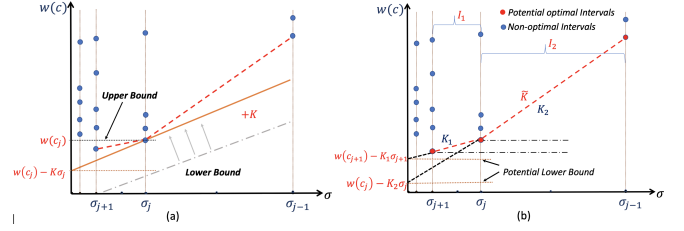


Fig. 3. (a) Fixed Lipschitz constant K ; (b) Adaptive Lipschitz constant $\tilde{K}$.

of the convex hull of the dots correspond to better function values with respect to the local search. We select all these intervals within the convex hull at each iteration. The adaptive Lipschitz constant $\tilde{K}$ can be calculated by the following equation.

$$\max_{i \in I_1} \frac{w(c_j) - w(c_i)}{\sigma_j - \sigma_i} \leq \tilde{K} \leq \min_{i \in I_2} \frac{w(c_i) - w(c_j)}{\sigma_i - \sigma_j} \quad (6)$$

Where I is the set of all indices of intervals, $I_1 = \{i \in I : \sigma_i < \sigma_j\}$, $I_2 = \{i \in I : \sigma_i > \sigma_j\}$, and j is the potentially optimal interval. In Fig. 3(b), we have $K_1 \leq \tilde{K} \leq K_2$, which also is the slope of the function value corresponding to the rectangle size.

2.3. Convergence and Complexity Analysis

For a continuous function, SORA would sample a dense subset of a unit hypercube. When the number of iterations tends to infinity, its convergence to the global minimum is guaranteed [23]. SORA only selects the potentially optimal space to evaluate. Based on Granham's Scan [31], the time complexity to find the convex hull of SORA is much smaller than the time complexity of finding the convex hull of a set of n arbitrary points in the grid search. The time complexity of finding the convex hull of a set of n arbitrary points is $O(n \log 2n)$. However, SORA has sorted the point by distance σ_j as shown in Fig. 3, the time complexity is only related to the distinct values of σ_j , denoted as n' , its time complexity is $O(n')$, $n' \ll n$. With a small tolerance ϵ after a finite number of queries, the convergence of SORA is guaranteed if the objective function is Lipschitz continuous. For a n dimensional input space C , its diameter is $d(C) = \max_{(c, c') \in C} \|c - c'\|_\infty$ with the Lipschitz constant K . After q queries, the gap between current minima and global minima is: $\min_{i=1 \dots q} w(c_i) - \min_{c \in C} w(c) \leq \epsilon \leq K \cdot d(C) \cdot 3^{-l}$, where l is the divided level of the current largest subspace in the unit search space. Assuming dividing one hyperrectangle in each query, after q query, there is no level of $(l - 1)$ hyperrectangle left, then $q = 3^{n-1}(3^{nl} - 1/3^n - 1) < 3^{nl} - 1$, so the relationship between q and l can be written as $l \geq \log_3(q + 1)/n$. Then we have

$$\epsilon \leq K \cdot d(C) \cdot 3^{-\frac{\log_3(q+1)}{n}} = K \cdot d(C)(q + 1)^{-\frac{1}{n}}.$$

Therefore, to achieve any small desired tolerance ϵ , the number of queries is given by $O((d(x) \cdot K/\epsilon)^n)$.

Table 1. Reachability analysis on MNIST classifiers. We report the average value of the minimum found in the first 100 examples in MNIST testset. Running time is up to 3 minutes.

Perturb. size	Methods	FCN (#Layer $\times$ #Unit)			CNN (#Total neuron)		
		6 $\times$ 200	9 $\times$ 200	6 $\times$ 500	5704	13,798	88,544
3 $\times$ 3	PGD	8.93	8.24	5.95	12.63	10.40	8.10
	DeepGO	8.97	8.31	5.99	12.63	10.33	8.10
	ERAN	9.71	9.12	6.49	13.24	13.52	8.64
	SORA	9.17	8.48	6.04	12.59	10.04	8.06
6 $\times$ 6	PGD	6.94	6.26	4.19	11.41	7.06	6.94
	DeepGO	timeout	timeout	timeout	timeout	timeout	timeout
	ERAN	9.71	9.12	6.49	13.24	13.52	8.64
	SORA	7.12	6.54	4.25	11.17	6.49	6.93
12 $\times$ 12	PGD	1.85	1.56	-5.18	6.03	1.46	2.51
	DeepGO	timeout	timeout	timeout	timeout	timeout	timeout
	ERAN	9.71	9.12	6.49	13.24	13.52	8.64
	SORA	4.61	4.38	0.48	9.18	2.74	4.32
28 $\times$ 28	PGD	1.13	1.54	-21.85	-2.16	0.51	-1.47
	DeepGO	timeout	timeout	timeout	timeout	timeout	timeout
	ERAN	-13297	-1475812	-17894	-77.09	-7990	timeout
	SORA	3.59	2.95	3.47	9.84	4.52	4.99

3. EXPERIMENTS

This section evaluates the performance of SORA and compares it with other robustness analysis methods. All experiments were performed on a workstation with 96GB RAM, i9-10900X CPU, and a 2080Ti GPU. First, we analyse SORA’s performance in analysing the reachability of DNN classifiers on the MNIST dataset. For comparison, we adopt DeepGO [24], PGD, a commonly used adversarial attack [28], and ERAN [12], a cutting-edge abstract interpretation-based robustness, as baseline methods. Since the input dimension is one potential limitation of DeepGO, we chose perturbation areas with different sizes to test how many dimensions that SORA can handle. Second, we use SORA to find the lower bound of the maximum safe norm ball (MSNB) $\arg \max_d \mathcal{B}(x, d, \infty)$ that makes Definition 1 hold. We run SORA, PGD [28], and $\alpha\beta$ -crown [13] repeatedly and perform a binary search to approximate the maximum safe norm ball. As explained in Section 2.1 and Fig.1, adversarial attack methods (e.g., PGD attack) essentially approximate the MSNB from its upper bound (between d_{max} and d_2), which lacks guarantees of certifying the nonexistence of adversarial examples. By contrast, verification methods (e.g., $\alpha\beta$ -crown, SORA) approximate the MSNB from its lower bound (between d_1 and d_{max}), which can certify the nonexistence of adversarial examples within the bounded region. Generally speaking, attacks aim to generate adversarial examples, while verification can prove non-existence of adversarial examples in a certain input region with provable guarantee. Verification on DNNs is essentially an NP-hard problem and much more challenging than attacks [32].

3.1. Comparison Results with Prior Work

Table 1 reports the average value of the minimum found in the first 100 examples of MNIST testset. Although PGD, as a

strong adversarial attack, can significantly reduce the model’s output toward the label, it is outperformed by SORA on complex CNNs when the perturbation areas are 3 $\times$ 3 and 6 $\times$ 6. Besides, SORA can analyse the whole example effectively, while DeepGO only works on 3 $\times$ 3 perturbation area. As for ERAN, it overestimates the output of the models and produces an impractical lower bound of the logit value, which is unsuitable for reachability analysis. It is worth mentioning that PGD and ERAN are white-box approaches that require full access to target models, while SORA is a query-based analyser that can work in a black-box environment. SORA overcomes the dimension limitation and can perform an effective readability analysis of any DNNs that are Lipschitz continuous.

On a pre-trained convolutional network with ReLU activations, we report the lower bound of MSNB on three randomly sampled examples from $\alpha\beta$ -crown and SORA, and the upper bound of the maximum safe norm ball is obtained from PGD. As shown in Table 2, we observe that PGD is efficient in terms of running time. However, it can only find an upper bound of the safe norm ball, which cannot verify whether the bounded region is safe. We also find that, for Example-9030, $\alpha\beta$ -crown produces an infeasible lower bound of MSNB since the bound is larger than the value from the PGD attack. In contrast, SORA can provide a valid lower bound. The empirical study also reveals that SORA can find a comparable lower bound of maximum safe norm ball using significantly less time than $\alpha\beta$ -crown on MNIST dataset, where PGD, $\alpha\beta$ -CROWN, and ERAN can only work in a white-box environment. SORA is a black-box, and standalone analyser that does not rely on adversarial attacks to compute the MSNB.

Table 2. Compare the maximum safe L_∞ -norm ball in the resized MNIST examples. The target model is a 4-layer CNNs with ReLU activation with accuracy 97.42% on testset.

Methods	Example Index			Avg.time
	9030	2999	5830	
PGD	0.018	0.041	0.039	37.7s
crown	0.024	0.036	0.037	50.3 mins
(without PGD)	(invalid)			
SORA	0.012	0.036	0.033	5.2 mins

4. CONCLUSION

In this paper, we propose a generic yet scalable reachability framework – SORA, which has provable guarantees and can be applied to large deep neural networks with very deep layers and nonlinear activation functions. The experiments demonstrate that our tool can achieve superior performance in terms of output range analysis and robustness verification. We envision that this work marks an important step toward a practical verification of DNNs. Future work includes parallelising this method on GPUs to test its scalability on large-scale models trained on ImageNet.

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